

Designing a Scalable IoT Operations SaaS Platform

A multi-tenant platform for water management, agritech, and air-quality operations.

ROLE

Lead UX/UI Designer

SCOPE

End-to-end UX, information architecture, role-based access, dashboards, design system, developer handoff

OUTCOME

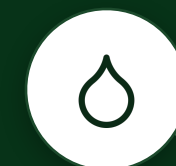
A scalable shared platform foundation supporting multiple IoT product lines

Aqua Ops

Hydrogrow

Air Intelligence

CONFIDENTIAL CASE STUDY



Water Monitoring



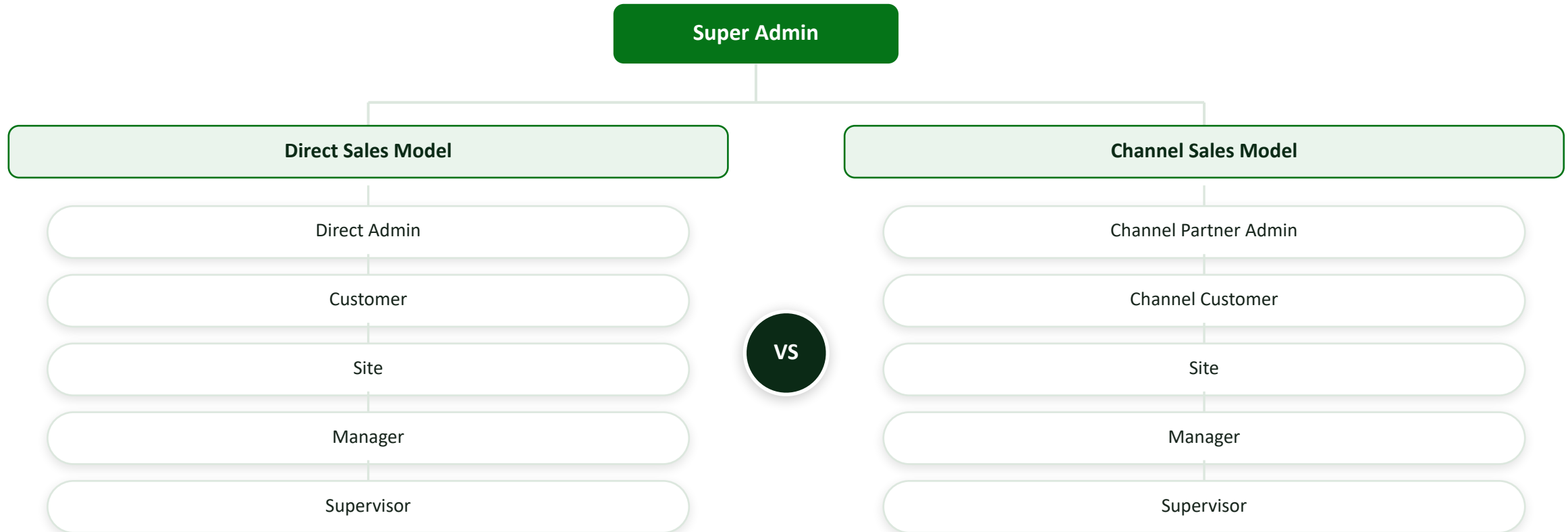
Agritech / Hydroponic



Air Quality Mgmt

THE CHALLENGE

One Platform, Multiple Business Models and User Hierarchies



Challenge: The platform needed to support different ownership models, role permissions, customers, sites, devices, and product branches — without creating operational complexity.

PLATFORM FOUNDATION

Designing a Reusable Multi-Tenant SaaS Foundation

CONFIGURABLE PRODUCT EXPERIENCES


Aqua Ops
 Water Quality Monitoring
 TDS · pH · Dissolved Oxygen · Device Performance
 Site Zones


Hydrogrow
 Crop & Nutrient Management
 EC · pH · Nutrient Levels · Crop Cycles · Grow Stage
 Crop Zones


Air Intelligence
 Environmental Air Monitoring
 CO₂ · PM2.5 · VOC · Humidity · Temperature
 Zone Context


User & Role Mgmt
 Shared


Customer & Channel
 Shared


Site Management
 Shared


Device Management
 Shared


Alerts & Controls
 Shared


Event & Audit Logs
 Shared

SCALABLE TENANT MODEL

Direct model: Super Admin → Direct Admin → Customer → Site → Manager → Supervisor

Channel model: Super Admin → Channel Partner Admin → Channel Customer → Site → Manager → Supervisor

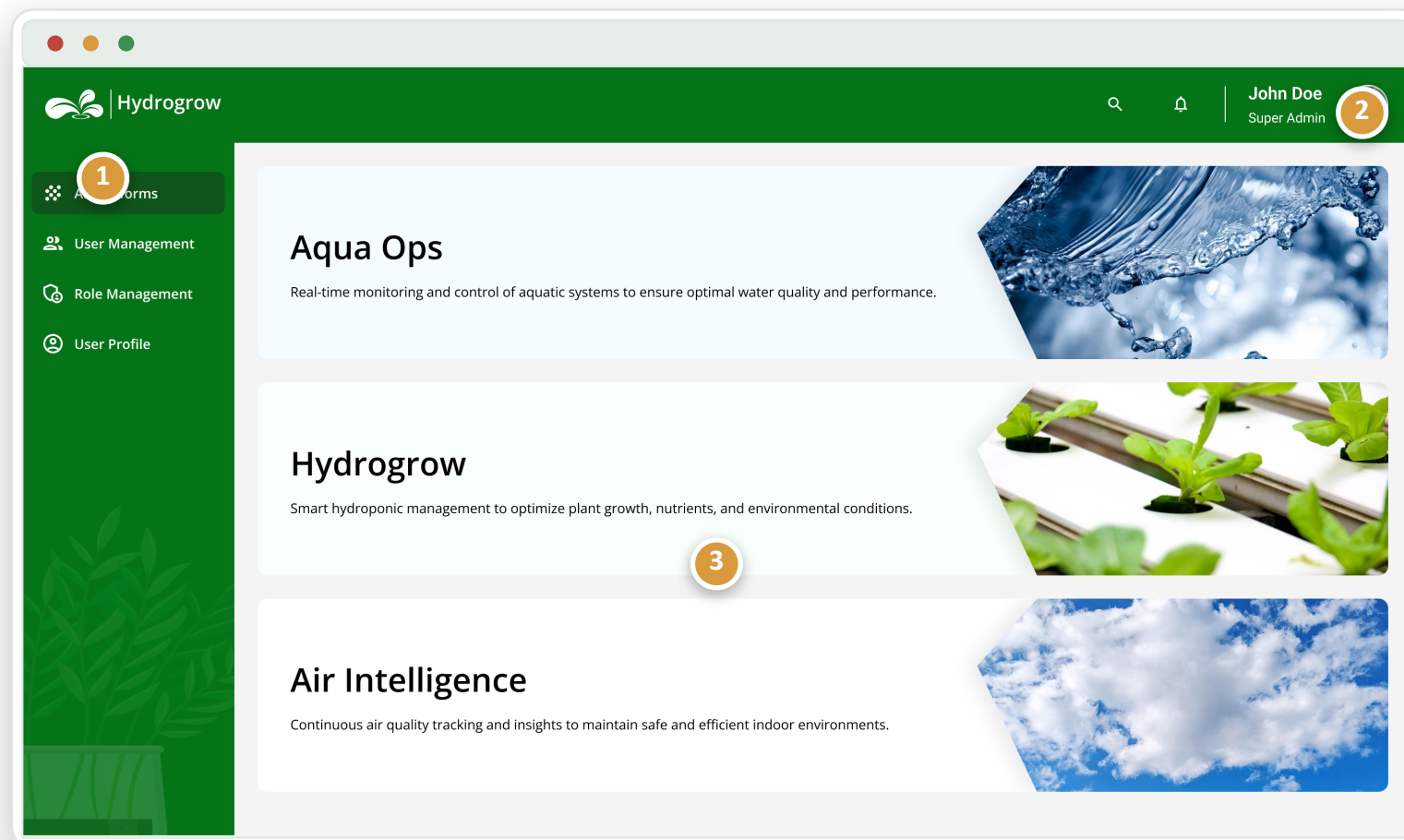
Designed once, scaled across three IoT businesses — a shared foundation for consistent administration, with each branch adapting its own devices, metrics, and workflows.

PRODUCT IN ACTION

One Platform, Three Product Experiences

Super Admins move between Aqua Ops, Hydrogrow, and Air Intelligence from a single unified shell — same session, same navigation pattern, per-product context.

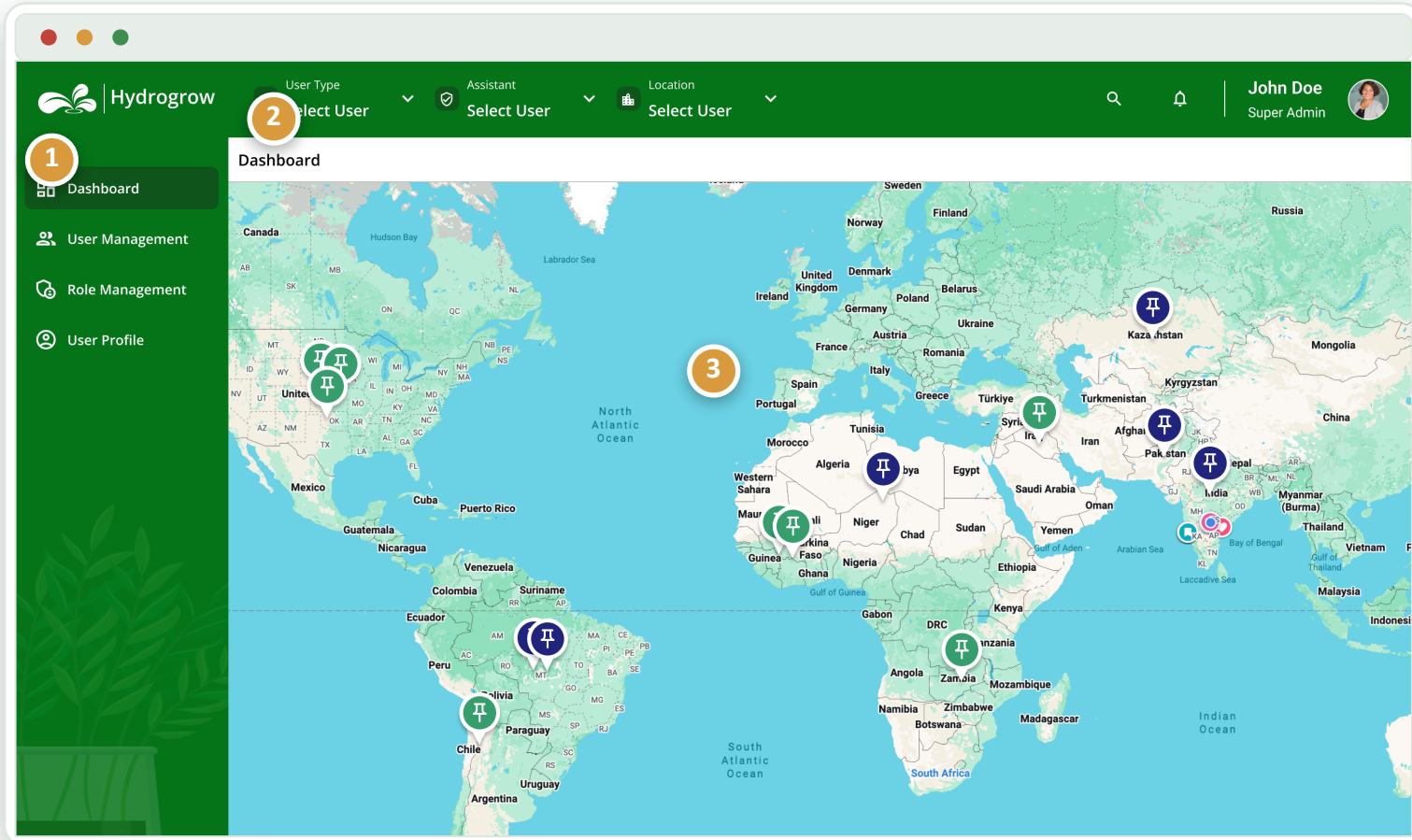
- Single sign-on across all three product lines
- Consistent nav shell and role context per product
- Instant switching without losing session state



1 Platform switcher

2 Super Admin session

3 Product cards



1 Dashboard nav

2 Role & location filters

3 Global device pins

PRODUCT IN ACTION

Global Operations at a Glance

A live, worldwide view of every site and device — filterable by user type, assistant, and location — turning distributed IoT operations into one governable dashboard.

- Real-time global site and device visibility
- Multi-level filters: user type, assistant, location
- Color-coded status pins for fast triage

X8 PLATFORM — HYDROGROW VERTICAL

A Conversational AI Layer for Vertical Farming

On the X8 platform, Hydrogrow with a chatbot layer so any user — farmer or layperson — can monitor and control IoT devices through simple voice or text commands.

BEFORE — OPERATIONAL CHALLENGES



Manual, time-consuming operations across sites



Water & energy resource inefficiency



Limited real-time monitoring



Complex, error-prone scheduling



Sensor data underused for insights

THE CHATBOT LAYER — KEY FEATURES



Remote Monitoring & Control

Check live temperature, humidity, and lighting through a simple chat interface.

"What's the humidity in Singapore 3, Layer System 1?"



Automated Scheduling

Set precise cycles for lights, fans, pumps, and watering — hands-free.

"Water the crops for 30 minutes every 6 hours."



Proactive Alerts & Maintenance

Get notified of malfunctions or threshold deviations before they escalate.

"Notify me when CO₂ exceeds 1200 ppm."

Results, Impact & What's Next

RESULTS & IMPACT

- ✓ Manual tasks — scheduling, reading system parameters — handled in one place.
- ✓ Users monitor real-time device performance and set their own alert thresholds.
- ✓ Any user, farmer or layperson, can manage tasks with minimal prior app or farming knowledge.

WHAT'S NEXT — ROADMAP



Real-Time Analysis & Forecasting

Predictive insights on water needs, crop cycles, and energy use from historical data.



Resource Optimization

AI-suggested efficient practices for water and energy, refined with harvest-cycle feedback.



Voice Assistant Integration

Siri, Alexa, and Google Assistant support for hands-free control and readings on the go.

Dependency: Roadmap features rely on structured feedback from farmers and MH after every harvest cycle — the model improves as real-world data accumulates.

Key UX Decisions

01



Role-Based Visibility

Each persona sees only the product areas, customers, sites, and devices relevant to their responsibilities.

02



Backend Device Provisioning

Devices are provisioned during manufacturing and identified by MAC ID through backend integration — not manually created or bulk imported.

03



Operational Traceability

Alert settings, event logs, and audit logs help Super Admins track device events and user actions over time.



Requirements

IA

Prototype

Validation

Hi-Fi UI

Dev Handoff

Contribution & Impact

Discovery & Research

Mapped user roles, workflows, and pain points across three product lines.



Design System

Built a reusable component library ensuring consistency across branches.



Delivery & Handoff

Documented specs and collaborated with engineering through implementation.



Architecture & IA

Designed scalable information architecture supporting the multi-tenant hierarchy.



Prototyping & Testing

Validated flows with stakeholders across all user roles.



Designed a platform foundation that scales across multiple IoT product verticals — enabling consistent UX while respecting each product's unique operational context.